

SOLIDWORKS API (.NET VBA C#) — Reference (179/337): IModelDoc2 (part 2 of 17)

SOLIDWORKS API Documentation (converted from CHM)

IModelDoc2 (part 2 of 17) (Type)

IModelDoc2 Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IModelDoc2 Interface

For a list of all members of this type, see [IModelDoc2 members](#).

 Public Properties

Name	Description
 ActiveView	Gets the current active model view in read-only mode. NOTE: This property is a get-only property. Set is not implemented.
 ConfigurationManager	Gets the IConfigurationManager object, which allows access to a configuration in a model.
 CustomInfo	Obsolete. Superseded by IModelDocExtension::CustomPropertyManager .
 CustomInfo2	Obsolete. Superseded by IModelDocExtension::CustomPropertyManager .
 Extension	Gets the IModelDocExtension object, which also allows access to the model document.
 FeatureManager	Gets the IFeatureManager object, which allows access to the FeatureManager design tree.
 FeatureManagerSplitterPosition	Splits the FeatureManager design tree and gets or sets the location of the split bar in the FeatureManager design tree panel.
 IActiveView	Gets the current active model view in read-only mode. NOTE: This property is a get-only property. Set is not implemented.
 ILightSourcePropertyValues	Gets and sets the light source property values.
 IMaterialPropertyValues	Gets or sets a material's properties in the active configuration.
 IPageSetup	Gets the page setup for this document.
 ISelectionManager	Gets the ISelectionMgr object for this document, which makes the currently selected object available.
 LargeAssemblyMode	Gets or sets Large Assembly Mode for this document.
 LengthUnit	Gets and sets the same LengthUnit value used by IModelDoc2::GetUnits , IModelDoc2::IGetUnits , and IModelDoc2::SetUnits .
 LightSourcePropertyValues	Gets and sets the light source property values.
 LightSourceUserName	Gets or sets the light source name that is displayed in the SOLIDWORKS Design user interface.
 MaterialIdName	Obsolete. Superseded by IPartDoc::GetMaterialPropertyName2 and IPartDoc::SetMaterialPropertyName2 .
 MaterialPropertyValues	Gets or sets a material's properties in the active configuration.
 MaterialUserName	Gets or sets the material name.
 ModelViewManager	Gets the IModelViewManager object, which allows access to the model view.
 PageSetup	Gets the page setup for this document.
 Printer	Gets or sets the default printer for this document.
 PrintSetup	Obsolete. See IModelDoc2::SetUserPreferenceDoubleValue , IModelDoc2::SetUserPreferenceIntegerValue , and IModelDoc2::SetUserPreferenceToggle .
 SceneBkgImageFileName	Controls the image filename used as the current background picture.
 SceneName	Gets and sets the name of the scene.
 SceneUserName	Gets and sets the user name of the scene.
 SelectionManager	Gets the ISelectionMgr object for this document, which makes the currently selected object available. NOTE: This property is a get-only property. Set is not implemented.
 ShowFeatureErrorDialog	Gets or sets whether to display the feature error dialog.
 SketchManager	Gets the sketch manager, which allows access to sketch-creation routines.
 SummaryInfo	Gets or sets file summary information for the SOLIDWORKS document.



Gets or sets the visibility of the active document.

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See Also

[IModelDoc2 Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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ActivateFeatureMgrView Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ActivateFeatureMgrView Method (IModelDoc2)

Obsolete. Superseded by [IFeatureMgrView::ActivateView](#) or [IModelDocExtension::ActivateFeatureMgrViewx64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

.NET Syntax

Visual Basic (Declaration)

```
Function ActivateFeatureMgrView( _  
    ByRef AppView As System.Integer _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim AppView As System.Integer  
Dim value As System.Integer  
  
value = instance.ActivateFeatureMgrView(AppView)
```

C#

```
System.int ActivateFeatureMgrView(  
    ref System.int AppView  
)
```

C++/CLI

```
System.int ActivateFeatureMgrView(  
& System.int% AppView  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AppView

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ActivateFeatureMgrView](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

See Also

ActivateSelectedFeature Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ActivateSelectedFeature Method (IModelDoc2)

Activates the selected feature for editing.

.NET Syntax

Visual Basic (Declaration)

```
Sub ActivateSelectedFeature()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.ActivateSelectedFeature()
```

C#

```
void ActivateSelectedFeature()
```

C++/CLI

```
void ActivateSelectedFeature();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ActivateSelectedFeature](#).

Example

[Activate Feature \(VBA\)](#)
[Create 3D Sketch Plane \(C#\)](#)
[Create 3D Sketch Plane \(VB.NET\)](#)
[Create 3D Sketch Plane \(VBA\)](#)

Remarks

This method shows the selected feature's dimensions in the graphic area.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

ActiveView Property (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ActiveView Property (IModelDoc2)

Gets the current active model view in read-only mode.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

Property ActiveView As System.Object

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As System.Object

instance.ActiveView = value

value = instance.ActiveView
```

C#

```
System.object ActiveView {get; set;}
```

C++/CLI

```
property System.Object^ ActiveView {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Current active [model view](#) in this document

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ActiveView](#).

Example

[Change Wrap Feature \(VBA\)](#)
[Get Display State \(VBA\)](#)
[Set Model Display \(VBA\)](#)
[Create Hidden Undo Object \(VBA\)](#)
[Create Hidden Undo Object \(VB.NET\)](#)
[Create Hidden Undo Object \(C#\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(C#\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VB.NET\)](#)
[Get Names of Components, Window Handles, and DIBSECTIONS \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IActiveView Property](#)
[IModelDoc2::EnumModelViews Method](#)
[IModelDoc2::GetFirstModelView Method](#)
[IModelDoc2::IGetFirstModelView Method](#)
[IModelView::GetNext Method](#)
[IModelView::IGetNext Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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AddConfiguration Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddConfiguration Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::AddConfiguration3](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub AddConfiguration( _
    ByVal Name As System.String, _
```

```

ByVal Comment As System.String, _
ByVal AlternateName As System.String, _
ByVal SuppressByDefault As System.Boolean, _
ByVal HideByDefault As System.Boolean, _
ByVal MinFeatureManager As System.Boolean, _
ByVal InheritProperties As System.Boolean, _
ByVal Flags As System.UInteger _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDoc2
Dim Name As System.String
Dim Comment As System.String
Dim AlternateName As System.String
Dim SuppressByDefault As System.Boolean
Dim HideByDefault As System.Boolean
Dim MinFeatureManager As System.Boolean
Dim InheritProperties As System.Boolean
Dim Flags As System.UInteger

instance.AddConfiguration(Name, Comment, AlternateName, SuppressByDefault, HideByDefault,
MinFeatureManager, InheritProperties, Flags)

```

C#

```

void AddConfiguration(
    System.string Name,
    System.string Comment,
    System.string AlternateName,
    System.bool SuppressByDefault,
    System.bool HideByDefault,
    System.bool MinFeatureManager,
    System.bool InheritProperties,
    System.uint Flags
)

```

C++/CLI

```

void AddConfiguration(
& System.String^ Name,
& System.String^ Comment,
& System.String^ AlternateName,
& System.bool SuppressByDefault,
& System.bool HideByDefault,
& System.bool MinFeatureManager,
& System.bool InheritProperties,
& System.uint Flags
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Comment

AlternateName

SuppressByDefault

HideByDefault

MinFeatureManager

InheritProperties

Flags

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddConfiguration](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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AddConfiguration2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddConfiguration2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::AddConfiguration3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddConfiguration2( _  
    ByVal Name As System.String, _  
    ByVal Comment As System.String, _  
    ByVal AlternateName As System.String, _  
    ByVal SuppressByDefault As System.Boolean, _  
    ByVal HideByDefault As System.Boolean, _  
    ByVal MinFeatureManager As System.Boolean, _  
    ByVal InheritProperties As System.Boolean, _  
    ByVal Flags As System.UInteger _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Name As System.String  
Dim Comment As System.String  
Dim AlternateName As System.String  
Dim SuppressByDefault As System.Boolean  
Dim HideByDefault As System.Boolean  
Dim MinFeatureManager As System.Boolean  
Dim InheritProperties As System.Boolean  
Dim Flags As System.UInteger  
Dim value As System.Boolean
```

```
value = instance.AddConfiguration2(Name, Comment, AlternateName, SuppressByDefault, HideByDefault,  
MinFeatureManager, InheritProperties, Flags)
```

C#

```
System.bool AddConfiguration2(  
    System.string Name,  
    System.string Comment,  
    System.string AlternateName,  
    System.bool SuppressByDefault,  
    System.bool HideByDefault,  
    System.bool MinFeatureManager,  
    System.bool InheritProperties,  
    System.uint Flags  
)
```

C++/CLI

```
System.bool AddConfiguration2(  
& System.String^ Name,  
& System.String^ Comment,  
& System.String^ AlternateName,  
& System.bool SuppressByDefault,  
& System.bool HideByDefault,  
& System.bool MinFeatureManager,  
& System.bool InheritProperties,  
& System.uint Flags  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Comment

AlternateName

SuppressByDefault

HideByDefault

MinFeatureManager

InheritProperties

Flags

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddConfiguration2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddConfiguration3 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddConfiguration3 Method (IModelDoc2)

Adds a new configuration to this model document.

.NET Syntax

Visual Basic (Declaration)

```
Function AddConfiguration3( _  
    ByVal Name As System.String, _  
    ByVal Comment As System.String, _  
    ByVal AlternateName As System.String, _  
    ByVal Options As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Name As System.String  
Dim Comment As System.String  
Dim AlternateName As System.String  
Dim Options As System.Integer  
Dim value As System.Object  
  
value = instance.AddConfiguration3(Name, Comment, AlternateName, Options)
```

C#

```
System.object AddConfiguration3(  
    System.string Name,  
    System.string Comment,  
    System.string AlternateName,  
    System.int Options  
)
```

C++/CLI

```
System.Object^ AddConfiguration3(  
& System.String^ Name,  
& System.String^ Comment,  
& System.String^ AlternateName,
```

```
& System.int Options  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Name of the new configuration

Comment

Comment displayed in Configuration Properties

AlternateName

Alternate configuration name; used if swConfigOption_UseAlternateName is set to true

Options

Combination of one or more configuration options as defined in [swConfigurationOptions2_e](#)

Return Value

Newly created [configuration](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddConfiguration3](#).

Example

[Add Configuration and Promote Child Components in BOM \(C#\)](#)
[Add Configuration and Promote Child Components in BOM \(VB.NET\)](#)
[Add Configuration and Promote Child Components in BOM \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetConfigurationByName Method](#)
[IModelDoc2::IGetConfigurationByName Method](#)
[IModelDoc2::DeleteConfiguration2 Method](#)
[IModelDoc2::EditConfiguration3 Method](#)
[IModelDoc2::GetConfigurationCount Method](#)
[IModelDoc2::GetConfigurationNames Method](#)
[IModelDoc2::IAddConfiguration3 Method](#)
[IModelDoc2::IGetConfigurationNames Method](#)
[IModelDoc2::ShowConfiguration2 Method](#)
[IConfiguration Interface](#)
[IConfigurationManager Interface](#)
[IModelDoc2::ConfigurationManager Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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AddCustomInfo Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddCustomInfo Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::CustomPropertyManager](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddCustomInfo( _  
    ByVal FieldName As System.String, _  
    ByVal FieldType As System.String, _  
    ByVal FieldValue As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim FieldName As System.String  
Dim FieldType As System.String  
Dim FieldValue As System.String
```

```
Dim value As System.Boolean
```

```
value = instance.AddCustomInfo(FieldName, FieldType, FieldValue)
```

C#

```
System.bool AddCustomInfo(  
    System.string FieldName,  
    System.string FieldType,  
    System.string FieldValue  
)
```

C++/CLI

```
System.bool AddCustomInfo(  
& System.String^ FieldName,  
& System.String^ FieldType,  
& System.String^ FieldValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FieldName

FieldType

FieldValue

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddCustomInfo](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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AddCustomInfo2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddCustomInfo2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::CustomPropertyManager](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddCustomInfo2( _  
    ByVal FieldName As System.String, _  
    ByVal FieldType As System.Integer, _  
    ByVal FieldValue As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim FieldName As System.String  
Dim FieldType As System.Integer  
Dim FieldValue As System.String  
Dim value As System.Boolean
```

```
value = instance.AddCustomInfo2(FieldName, FieldType, FieldValue)
```

C#

```
System.bool AddCustomInfo2(  
    System.string FieldName,
```

```
System.int FieldType,  
System.string FieldValue  
)
```

C++/CLI

```
System.bool AddCustomInfo2(  
& System.String^ FieldName,  
& System.int FieldType,  
& System.String^ FieldValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FieldName

FieldType

FieldValue

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddCustomInfo2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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AddCustomInfo3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddCustomInfo3 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::CustomPropertyManager](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddCustomInfo3( _  
    ByVal Configuration As System.String, _  
    ByVal FieldName As System.String, _  
    ByVal FieldType As System.Integer, _  
    ByVal FieldValue As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Configuration As System.String  
Dim FieldName As System.String  
Dim FieldType As System.Integer  
Dim FieldValue As System.String  
Dim value As System.Boolean
```

```
value = instance.AddCustomInfo3(Configuration, FieldName, FieldType, FieldValue)
```

C#

```
System.bool AddCustomInfo3(  
    System.string Configuration,  
    System.string FieldName,  
    System.int FieldType,  
    System.string FieldValue  
)
```

C++/CLI

```
System.bool AddCustomInfo3(
& System.String^ Configuration,
& System.String^ FieldName,
& System.int FieldType,
& System.String^ FieldValue
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Configuration

FieldName

FieldType

FieldValue

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddCustomInfo3](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

AddDiameterDimension Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddDiameterDimension Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::AddDiameterDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddDiameterDimension( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As System.Boolean
```

```
value = instance.AddDiameterDimension(X, Y, Z)
```

C#

```
System.bool AddDiameterDimension(
    System.double X,
    System.double Y,
    System.double Z
)
```

C++/CLI

```
System.bool AddDiameterDimension(
& System.double X,
& System.double Y,
```

```
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddDiameterDimension](#).

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

SOLIDWORKS API Help

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AddDiameterDimension2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddDiameterDimension2 Method (IModelDoc2)

Adds a diameter dimension at the specified location for the selected item.

.NET Syntax

Visual Basic (Declaration)

```
Function AddDiameterDimension2( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object
```

```
value = instance.AddDiameterDimension2(X, Y, Z)
```

C#

```
System.object AddDiameterDimension2(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ AddDiameterDimension2(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

X location for the dimension

Y

Y location for the dimension

Z

Z location for the dimension

Return Value

Newly created diameter [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddDiameterDimension2](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDocExtension::AddSymmetricDimension Method](#)
[IModelDocExtension::AddSpecificDimension Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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AddDimension Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddDimension Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::AddDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddDimension( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean  
  
value = instance.AddDimension(X, Y, Z)
```

C#

```
System.bool AddDimension(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool AddDimension(  
& System.double X,
```

```
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddDimension](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

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AddDimension2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddDimension2 Method (IModelDoc2)

Creates a display dimension at the specified location for selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Function AddDimension2(  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.AddDimension2(X, Y, Z)
```

C#

```
System.object AddDimension2(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ AddDimension2(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

Dimension text location in meters

Y
Dimension text location in meters
Z
Dimension text location in meters

Return Value

Newly created [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddDimension2](#).

Remarks

Before calling this method, you must select the entities for which to add a display dimension.

If extension lines are needed during dimensioning because the selected entities do not unambiguously define what needs to be dimensioned, call [IModelDocExtension::AddDimension](#) instead of this method.

To dimension, make selections based on location and not name. For example, creating an angular dimension between two lines gets different results based on which line endpoints are selected. Therefore, if you are using [IModelDocExtension::SelectByID2](#) to select the entities for dimensioning, then you should specify the X, Y, and Z selection coordinates. You must also leave the object name argument empty, because if you pass the object name to IModelDocExtension::SelectByID2, then the selection routines use the object name instead of the coordinates to locate that item.

NOTE: To change the angle in an angular dimension to its supplementary angle, use [IDisplayDimension::SupplementaryAngle](#).

Because coordinates are ignored when the object name is passed, the dimensioning routines are not able to use a specific selection location for the dimension. This causes unpredictable results during dimension creation. IModelDocExtension::SelectByID2 randomly selects the line endpoint.

You should only use this method on visible documents. Before using this method, check to see if the document is visible by using [ISldWorks::Visible](#).

When you use this method, you might also want to use [ISldWorks::SetUserPreferenceToggle](#) with swInputDimValOnCreate to suppress the dialog box that lets the user type a dimension value.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::IAddDimension2 Method](#)
[IModelDoc2::IAddDiameterDimension2 Method](#)
[IModelDoc2::IAddHorizontalDimension2 Method](#)
[IModelDoc2::IAddRadialDimension2 Method](#)
[IModelDoc2::IAddVerticalDimension2 Method](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDocExtension::AddSpecificDimension Method \(\)](#)
[IDrawingDoc::InsertChainDim Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddFeatureMgrView Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddFeatureMgrView Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::AddFeatureMgrView3x64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

.NET Syntax

Visual Basic (Declaration)

```
Function AddFeatureMgrView( _  
    ByRef Bitmap As System.Integer, _  
    ByRef AppView As System.Integer _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Bitmap As System.Integer  
Dim AppView As System.Integer  
Dim value As System.Boolean  
  
value = instance.AddFeatureMgrView(Bitmap, AppView)
```

C#

```
System.bool AddFeatureMgrView(  
    ref System.int Bitmap,  
    ref System.int AppView  
)
```

C++/CLI

```
System.bool AddFeatureMgrView(  
& System.int% Bitmap,  
& System.int% AppView  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

AppView

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddFeatureMgrView](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddFeatureMgrView2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddFeatureMgrView2 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::AddFeatureMgrView3x64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

.NET Syntax

Visual Basic (Declaration)

```
Function AddFeatureMgrView2( _
    ByRef Bitmap As System.Integer, _
    ByRef AppView As System.Integer, _
    ByVal ToolTip As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim Bitmap As System.Integer
Dim AppView As System.Integer
Dim ToolTip As System.String
Dim value As System.Boolean

value = instance.AddFeatureMgrView2(Bitmap, AppView, ToolTip)
```

C#

```
System.bool AddFeatureMgrView2(
    ref System.int Bitmap,
    ref System.int AppView,
    System.string ToolTip
)
```

C++/CLI

```
System.bool AddFeatureMgrView2(
    & System.int% Bitmap,
    & System.int% AppView,
    & System.String^ ToolTip
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

AppView

ToolTip

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddFeatureMgrView2](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddFeatureMgrView3 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddFeatureMgrView3 Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::AddFeatureMgrView3x64](#).

This method is unsafe to use with 64-bit SOLIDWORKS, i.e. all versions since SOLIDWORKS 2016.

The 32-bit integer returned may be too small to contain the 64-bit window handle/pointer, and will be truncated silently. A critical exception will occur sometime later, when the truncated handle/pointer is used.

Adds the specified tab to the FeatureManager design tree view.

.NET Syntax

Visual Basic (Declaration)

```
Function AddFeatureMgrView3( _  
    ByRef Bitmap As System.Integer, _  
    ByRef AppView As System.Integer, _  
    ByVal ToolTip As System.String, _  
    ByVal WhichPane As System.Integer _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Bitmap As System.Integer  
Dim AppView As System.Integer  
Dim ToolTip As System.String  
Dim WhichPane As System.Integer  
Dim value As System.Boolean  
  
value = instance.AddFeatureMgrView3(Bitmap, AppView, ToolTip, WhichPane)
```

C#

```
System.bool AddFeatureMgrView3(  
    ref System.int Bitmap,  
    ref System.int AppView,  
    System.string ToolTip,  
    System.int WhichPane  
)
```

C++/CLI

```
System.bool AddFeatureMgrView3(  
& System.int% Bitmap,  
& System.int% AppView,  
& System.String^ ToolTip,  
& System.int WhichPane  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Bitmap

Pointer to the bitmap that you want to use for the FeatureManager design tree tab; the bitmap should be no larger than 16x18; this standard pointer is created by performing a New on Cbitmap

AppView

Pointer to an existing view that you want to use as your FeatureManager design tree view. If you do not have an existing view, then use [IModelViewMgr::CreateFeatureMgrView2](#), which returns a view handle

ToolTip

ToolTip string

WhichPane

Pane in which to add the view as defined in [swFeatMgrPane_e](#)

Return Value

True if the FeatureManager design tree view tab is added successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddFeatureMgrView3](#).

Remarks

All applications are required to be compatible with SOLIDWORKS x64. You must replace all calls to 32-bit only APIs with calls to their 64-bit safe counterparts.

See [Eliminating 32-bit only API use](#) for more information.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDocExtension::AddFeatureMgrView3x64 Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddHorizontalDimension Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddHorizontalDimension Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::AddHorizontalDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddHorizontalDimension( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean  
  
value = instance.AddHorizontalDimension(X, Y, Z)
```

C#

```
System.bool AddHorizontalDimension(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool AddHorizontalDimension(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddHorizontalDimension](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

AddHorizontalDimension2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddHorizontalDimension2 Method (IModelDoc2)

Creates a horizontal dimension for the currently selected entities at the specified location.

.NET Syntax

Visual Basic (Declaration)

```
Function AddHorizontalDimension2( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.AddHorizontalDimension2(X, Y, Z)
```

C#

```
System.object AddHorizontalDimension2(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ AddHorizontalDimension2(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

[X](#)
Dimension text location in meters

[Y](#)
Dimension text location in meters

[Z](#)
Dimension text location in meters

Return Value

Newly created [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddHorizontalDimension2](#).

Example

[Dimension Edge in Drawing \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDocExtension::AddSymmetricDimension Method](#)
[IModelDocExtension::AddSpecificDimension Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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AddIns Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddIns Method (IModelDoc2)

Displays the **Add-In Manager** dialog box.

.NET Syntax

Visual Basic (Declaration)

```
Sub AddIns()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.AddIns()
```

C#

```
void AddIns()
```

C++/CLI

```
void AddIns();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddIns](#).

Remarks

See [ISldWorks::LoadAddIn](#) for details on how to programmatically load an add-in.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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AddLightSource Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddLightSource Method (IModelDoc2)

Adds a type of light source to a scene with the specified names.

.NET Syntax

Visual Basic (Declaration)

```
Function AddLightSource( _  
    ByVal IdName As System.String, _  
    ByVal LTyp As System.Integer, _  
    ByVal UserName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim IdName As System.String  
Dim LTyp As System.Integer  
Dim UserName As System.String  
Dim value As System.Boolean  
  
value = instance.AddLightSource(IdName, LTyp, UserName)
```

C#

```
System.bool AddLightSource(  
    System.string IdName,  
    System.int LTyp,  
    System.string UserName  
)
```

C++/CLI

```
System.bool AddLightSource(  
& System.String^ IdName,  
& System.int LTyp,  
& System.String^ UserName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

IdName

New light source ID name

LTyp

New light source type:

- 1 = Ambient light
- 2 = Spot light
- 3 = Point light
- 4 = Directional light

UserName

New light source user name

Return Value

True if the light source creation was successful, false if not; for example, creation of a light source fails if IdName is not unique within this model

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddLightSource](#).

Example

[Add Spotlight and Get Light Feature \(C#\)](#)
[Add Spotlight and Get Light Feature \(VB.NET\)](#)
[Add Spotlight and Get Light Feature \(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddLightSourceExtProperty Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::GetLightSourceCount Method](#)
[IModelDoc2::GetLightSourceExtProperty Method](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::ResetLightSourceExtProperty Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)
[IModelDoc2::ILightSourcePropertyValues Property](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourceUserName Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddLightSourceExtProperty Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddLightSourceExtProperty Method (IModelDoc2)

Stores a float, string, or integer value for the light source.

.NET Syntax

Visual Basic (Declaration)

```
Function AddLightSourceExtProperty( _  
    ByVal ID As System.Integer, _  
    ByVal PropertyExtension As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim ID As System.Integer  
Dim PropertyExtension As System.Object  
Dim value As System.Integer  
  
value = instance.AddLightSourceExtProperty(ID, PropertyExtension)
```

C#

```
System.int AddLightSourceExtProperty(  
    System.int ID,  
    System.object PropertyExtension  
)
```

C++/CLI

```
System.int AddLightSourceExtProperty(  
& System.int ID,  
& System.Object^ PropertyExtension  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
ID of the light source
PropertyExtension
Value you want to store for the light source

Return Value

ID of the extension property (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddLightSourceExtProperty](#).

Remarks

This property extension is stored on the model document and is unique to the specified light source.

To add the property extension, you must first define the VARIANT type (float, string, or integer), give your variable a value, and then call this method to place the value on the light source for future reference.

This method returns a 1-based index of the added property. Values returned from `IModelDoc2::AddLightSourceExtProperty` must be decremented by 1 when used as input to [IModelDoc2::GetLightSourceExtProperty](#). However, the error value of -1 should not be decremented.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddLightSource Method](#)
[IModelDoc2::DeleteLightSource Method](#)
[IModelDoc2::GetLightSourceExtProperty Method](#)
[IModelDoc2::GetLightSourceIdFromName Method](#)
[IModelDoc2::GetLightSourceName Method](#)
[IModelDoc2::ResetLightSourceExtProperty Method](#)
[IModelDoc2::SetLightSourceName Method](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method](#)
[IModelDoc2::ILightSourcePropertyValues Property](#)
[IModelDoc2::LightSourcePropertyValues Property](#)
[IModelDoc2::LightSourceUserName Property](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddLightToScene Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddLightToScene Method (IModelDoc2)

Adds a light source to a scene.

.NET Syntax

Visual Basic (Declaration)

```
Function AddLightToScene( _  
    ByVal LpszNewValue As System.String _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim LpszNewValue As System.String  
Dim value As System.Integer  
  
value = instance.AddLightToScene(LpszNewValue)
```

C#

```
System.int AddLightToScene(  
    System.string LpszNewValue  
)
```

C++/CLI

```
System.int AddLightToScene(  
& System.String^ LpszNewValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

LpszNewValue

Name to use for the light

Return Value

ID of the light

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddLightToScene](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddLoftSection Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddLoftSection Method (IModelDoc2)

Adds a loft section to a blend feature.

.NET Syntax

Visual Basic (Declaration)

```
Sub AddLoftSection()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.AddLoftSection()
```

C#

```
void AddLoftSection()
```

C++/CLI

```
void AddLoftSection();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddLoftSection](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddOrEditConfiguration Method (IModelDoc2)

[See Also](#)



Obsolete. Superseded by [IConfiguration::GetParameters](#), [IConfiguration::IGetParameters](#), [IConfiguration::ISetParameters](#), and [IConfiguration::SetParameters](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddOrEditConfiguration( _
    ByVal ConfigName As System.String, _
    ByVal Params As System.Object, _
    ByVal Values As System.Object _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim ConfigName As System.String
Dim Params As System.Object
Dim Values As System.Object
Dim value As System.Integer

value = instance.AddOrEditConfiguration(ConfigName, Params, Values)
```

C#

```
System.int AddOrEditConfiguration(
    System.string ConfigName,
    System.object Params,
    System.object Values
)
```

C++/CLI

```
System.int AddOrEditConfiguration(
    & System.String^ ConfigName,
    & System.Object^ Params,
    & System.Object^ Values
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigName

Params

Values

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddOrEditConfiguration](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

AddPropertyExtension Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddPropertyExtension Method (IModelDoc2)

Adds a property extension to this model.

.NET Syntax

Visual Basic (Declaration)

```
Function AddPropertyExtension( _  
    ByVal PropertyExtension As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim PropertyExtension As System.Object  
Dim value As System.Integer  
  
value = instance.AddPropertyExtension(PropertyExtension)
```

C#

```
System.int AddPropertyExtension(  
    System.object PropertyExtension  
)
```

C++/CLI

```
System.int AddPropertyExtension(  
& System.Object^ PropertyExtension  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropertyExtension

Value of the property extension to add to the model (see **Remarks**)

Return Value

Size of the array to which the property extension value is added

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddPropertyExtension](#).

Remarks

To use this method:

1. Declare the variable.
2. Assign the variable a value: float, integer, or string.
3. Call this method to add the value to the model.

The 1-based array is a first-in-last-out structured list, whose size is used by [IModelDoc2::GetPropertyExtension](#) to access the property extension..

NOTE: SOLIDWORKS recommends that you use [IAttribute](#), [IAttributeDef](#), and [IParameter](#) instead of this method. These interfaces provide more flexibility.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::ResetPropertyExtension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddRadialDimension Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddRadialDimension Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::AddRadialDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddRadialDimension( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean  
  
value = instance.AddRadialDimension(X, Y, Z)
```

C#

```
System.bool AddRadialDimension(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool AddRadialDimension(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddRadialDimension](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddRadialDimension2 Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddRadialDimension2 Method (IModelDoc2)

Adds a radial dimension at the specified location for the selected item.

.NET Syntax

Visual Basic (Declaration)

```
Function AddRadialDimension2( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _
```

```
ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object  
  
value = instance.AddRadialDimension2(X, Y, Z)
```

C#

```
System.object AddRadialDimension2(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.Object^ AddRadialDimension2(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
Dimension text location in meters

Y
Dimension text location in meters

Z
Dimension text location in meters

Return Value

[Dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddRadialDimension2](#).

Remarks

You can only create radial dimensions on sketch segments in an active sketch.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddVerticalDimension2 Method](#)
[IModelDoc2::IAddDiameterDimension2 Method](#)
[IModelDoc2::IAddDimension2 Method](#)
[IModelDoc2::IAddHorizontalDimension2 Method](#)
[IModelDoc2::IAddRadialDimension2 Method](#)
[IModelDoc2::IAddVerticalDimension2 Method](#)
[IModelDocExtension::AddSymmetricDimension Method](#)
[IDisplayDimension::ArcExtensionLineOrOppositeSide Property](#)
[IModelDocExtension::AddSpecificDimension Method \(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

AddRelation Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddRelation Method (IModelDoc2)

Obsolete. Superseded by [IEquationMgr::Add](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub AddRelation( _  
    ByVal RelStr As System.String _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim RelStr As System.String  
  
instance.AddRelation(RelStr)
```

C#

```
void AddRelation(  
    System.string RelStr  
)
```

C++/CLI

```
void AddRelation(  
& System.String^ RelStr  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

RelStr

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddRelation](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

AddSceneExtProperty Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddSceneExtProperty Method (IModelDoc2)

Stores a float, string, or integer value for a scene.

.NET Syntax

Visual Basic (Declaration)

```
Function AddSceneExtProperty(  
    ByVal PropertyExtension As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim PropertyExtension As System.Object
```

```
Dim value As System.Integer

value = instance.AddSceneExtProperty(PropertyExtension)
```

C#

```
System.int AddSceneExtProperty(
    System.object PropertyExtension
)
```

C++/CLI

```
System.int AddSceneExtProperty(
    & System.Object^ PropertyExtension
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PropertyExtension
Value for the scene

Return Value

Unique identifier returned to access the property extension in the future

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddSceneExtProperty](#).

Remarks

This scene extension property is stored on the model document, but is unique to the model's scene. To add the extension property, you must first define the VARIANT type (float, string, or integer), give your variable a value, and then call this method to place the value on the light source for future reference.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AddVerticalDimension Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddVerticalDimension Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::AddVerticalDimension2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AddVerticalDimension( _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDoc2
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As System.Boolean
```



```
value = instance.AddVerticalDimension(X, Y, Z)
```

C#

```
System.bool AddVerticalDimension(  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool AddVerticalDimension(  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddVerticalDimension](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

AddVerticalDimension2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AddVerticalDimension2 Method (IModelDoc2)

Creates a vertical dimension for the currently selected entities at the specified location.

.NET Syntax

Visual Basic (Declaration)

```
Function AddVerticalDimension2( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Object
```

```
value = instance.AddVerticalDimension2(X, Y, Z)
```

C#

```
System.object AddVerticalDimension2(  
    System.double X,  
    System.double Y,
```

```
        System.double Z  
    )
```

C++/CLI

```
System.Object^ AddVerticalDimension2(  
    & System.double X,  
    & System.double Y,  
    & System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
Dimension text location in meters

Y
Dimension text location in meters

Z
Dimension text location in meters

Return Value

Newly created [dimension](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AddVerticalDimension2](#).

Example

[Dimension Edge in Drawing_\(VBA\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AddDimension2 Method](#)
[IModelDoc2::AddHorizontalDimension2 Method](#)
[IModelDoc2::AddDiameterDimension2 Method](#)
[IModelDoc2::AddRadialDimension2 Method](#)
[IModelDoc2::IAddDiameterDimension2 Method](#)
[IModelDoc2::IAddDimension2 Method](#)
[IModelDoc2::IAddHorizontalDimension2 Method](#)
[IModelDoc2::IAddRadialDimension2 Method](#)
[IModelDoc2::IAddVerticalDimension2 Method](#)
[IModelDocExtension::AddSymmetricDimension Method](#)
[IModelDocExtension::AddSpecificDimension Method\(\)](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

AlignDimensions Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AlignDimensions Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::AlignDimensions](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub AlignDimensions()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.AlignDimensions()
```

C#

```
void AlignDimensions()
```

C++/CLI

```
void AlignDimensions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AlignDimensions](#).

Remarks

This method attempts to sort the selected dimensions into three groups: linear, ordinate, and angular. Within the linear group, it sorts by measured direction. Each of these dimensions are then aligned with the other like dimensions. These are then updated and dragged together.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AlignOrdinate Method](#)
[IModelDoc2::AlignParallelDimensions Method](#)
[IModelDoc2::BreakDimensionAlignment Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AlignOrdinate Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AlignOrdinate Method (IModelDoc2)

Aligns the selected group of ordinate dimensions.

.NET Syntax

Visual Basic (Declaration)

```
Sub AlignOrdinate()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.AlignOrdinate()
```

C#

```
void AlignOrdinate()
```

C++/CLI

```
void AlignOrdinate();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AlignOrdinate](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AlignDimensions Method](#)
[IModelDoc2::AlignParallelDimensions Method](#)
[IModelDocExtension::AddOrdinateDimension Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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AlignParallelDimensions Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AlignParallelDimensions Method (IModelDoc2)

Aligns the selected linear dimensions in a parallel fashion.

.NET Syntax

Visual Basic (Declaration)

```
Sub AlignParallelDimensions()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.AlignParallelDimensions()
```

C#

```
void AlignParallelDimensions()
```

C++/CLI

```
void AlignParallelDimensions();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AlignParallelDimensions](#).

Remarks

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::AlignDimensions Method](#)
[IModelDoc2::AlignOrdinate Method](#)
[IModelDoc2::AlignParallelDimensions Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

AndSelect Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AndSelect Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::SelectByID2](#).

.NET Syntax

```
Sub AndSelect( _
    ByVal SelID As System.String, _
    ByVal SelParams As System.String, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double _
)
```

```
Dim instance As IModelDoc2
Dim SelID As System.String
Dim SelParams As System.String
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double

instance.AndSelect(SelID, SelParams, X, Y, Z)
```

```
void AndSelect(
    System.string SelID,
    System.string SelParams,
    System.double X,
    System.double Y,
    System.double Z
)
```

```
void AndSelect(
    & System.String^ SelID,
    & System.String^ SelParams,
    & System.double X,
    & System.double Y,
    & System.double Z
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SelID

SelParams

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AndSelect](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

AndSelectByID Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AndSelectByID Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::SelectByID2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AndSelectByID( _  
    ByVal SelID As System.String, _  
    ByVal SelParams As System.String, _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double, _  
    ByVal Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim SelID As System.String  
Dim SelParams As System.String  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim value As System.Boolean
```

```
value = instance.AndSelectByID(SelID, SelParams, X, Y, Z)
```

C#

```
System.bool AndSelectByID(  
    System.string SelID,  
    System.string SelParams,  
    System.double X,  
    System.double Y,  
    System.double Z  
)
```

C++/CLI

```
System.bool AndSelectByID(  
& System.String^ SelID,  
& System.String^ SelParams,  
& System.double X,  
& System.double Y,  
& System.double Z  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SelID

SelParams

X

Y

Z

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AndSelectByID](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Obsolete. Superseded by [IModelDocExtension::SelectByID2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function AndSelectByMark( _
    ByVal SelID As System.String, _
    ByVal SelParams As System.String, _
    ByVal X As System.Double, _
    ByVal Y As System.Double, _
    ByVal Z As System.Double, _
    ByVal Mark As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim SelID As System.String
Dim SelParams As System.String
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim Mark As System.Integer
Dim value As System.Boolean

value = instance.AndSelectByMark(SelID, SelParams, X, Y, Z, Mark)
```

C#

```
System.bool AndSelectByMark(
    System.string SelID,
    System.string SelParams,
    System.double X,
    System.double Y,
    System.double Z,
    System.int Mark
)
```

C++/CLI

```
System.bool AndSelectByMark(
    & System.String^ SelID,
    & System.String^ SelParams,
    & System.double X,
    & System.double Y,
    & System.double Z,
    & System.int Mark
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SelID

SelParams

X

Y

Z

Mark

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AndSelectByMark](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

AutoInferToggle Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AutoInferToggle Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::AutoInference](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub AutoInferToggle()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.AutoInferToggle()
```

C#

```
void AutoInferToggle()
```

C++/CLI

```
void AutoInferToggle();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AutoInferToggle](#).

Remarks

Inferencing mode can be seen when creating a sketch segment and you mouse moves past another sketch item. If inferencing is turned on, you see a dashed line from the current cursor position to the inferred position on the existing sketch entity.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::SetInferenceMode Method](#)
[IModelDoc2::GetInferenceMode Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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AutoSolveToggle Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : AutoSolveToggle Method (IModelDoc2)

Obsolete. Superseded by [ISketchManager::AutoSolve](#).

.NET Syntax

Visual Basic (Declaration)

Sub AutoSolveToggle()

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.AutoSolveToggle()
```

C#

```
void AutoSolveToggle()
```

C++/CLI

```
void AutoSolveToggle();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::AutoSolveToggle](#).

Remarks

If this method is set on, then computations are solved automatically. When you are in the process of changing many dimensions in an active sketch, you may want to temporarily turn off automatic solving.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

BlankRefGeom Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : BlankRefGeom Method (IModelDoc2)

Hides the selected reference geometry in the graphics window.

.NET Syntax

Visual Basic (Declaration)

```
Sub BlankRefGeom()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.BlankRefGeom()
```

C#

```
void BlankRefGeom()
```

C++/CLI

```
void BlankRefGeom();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::BlankRefGeom](#).

 Example

[Create Replace Face Feature \(C#\)](#)
[Create Replace Face Feature \(VB.NET\)](#)
[Create Replace Face Feature \(VBA\)](#)

 Remarks

If you select a reference plane and then call this method, the plane is invisible in the graphics area. This method has the same effect as right-clicking an item and selecting **Hide** on the shortcut menu.

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::UnBlankRefGeom Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

BlankSketch Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : BlankSketch Method (IModelDoc2)

Hides the selected sketches.

 .NET Syntax

Visual Basic (Declaration)

Sub BlankSketch()

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.BlankSketch()
```

C#

```
void BlankSketch()
```

C++/CLI

```
void BlankSketch();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::BlankSketch](#).

 Example

[Traverse Assembly and Hide All Sketches \(VBA\)](#)
[Reset Visibility of Sketches in Drawing View \(C#\)](#)
[Reset Visibility of Sketches in Drawing View \(VB.NET\)](#)
[Reset Visibility of Sketches in Drawing View \(VBA\)](#)
[Hide and Show Sketches \(C#\)](#)
[Hide and Show Sketches \(VB.NET\)](#)
[Hide and Show Sketches \(VBA\)](#)

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::UnblankSketch Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

BreakAllExternalReferences Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : BreakAllExternalReferences Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::BreakAllExternalReferences2](#)

 .NET Syntax

Visual Basic (Declaration)

```
Sub BreakAllExternalReferences()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.BreakAllExternalReferences()
```

C#

```
void BreakAllExternalReferences()
```

C++/CLI

```
void BreakAllExternalReferences();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

 Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::BreakAllExternalReferences](#).

 See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetExternalReferenceName Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)
[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)
[IModelDoc2::LockAllExternalReferences Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::ListExternalFileReferencesCount Method](#)
[IModelDocExtension::ListExternalFileReferences Method](#)
[IModelDocExtension::UpdateExternalFileReferences Method](#)

 Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

BreakDimensionAlignment Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : BreakDimensionAlignment Method (IModelDoc2)

Breaks the association of any selected dimensions belonging to an alignment group (parallel or collinear).

 .NET Syntax

Visual Basic (Declaration)

```
Sub BreakDimensionAlignment()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.BreakDimensionAlignment()
```

C#

```
void BreakDimensionAlignment()
```

C++/CLI

```
void BreakDimensionAlignment();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::BreakDimensionAlignment](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ChangeSketchPlane Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ChangeSketchPlane Method (IModelDoc2)

Obsolete. Superseded by [IModelDocExtension::ChangeSketchPlane](#).

.NET Syntax

Visual Basic (Declaration)

```
Function ChangeSketchPlane() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim value As System.Boolean  
  
value = instance.ChangeSketchPlane()
```

C#

```
System.bool ChangeSketchPlane()
```

C++/CLI

```
System.bool ChangeSketchPlane();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the change was successful, false otherwise

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ChangeSketchPlane](#).

Example

[Change Plane of Sketch \(VBA\)](#)

Remarks

Every sketch is associated with a plane (for example, a reference plane or a planar face). You must preselect the sketch and the new plane or face before using this method.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ClearSelection Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ClearSelection Method (IModelDoc2)

Obsolete. Superseded by [IModelDoc2::ClearSelection2](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub ClearSelection()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
```

```
instance.ClearSelection()
```

C#

```
void ClearSelection()
```

C++/CLI

```
void ClearSelection();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ClearSelection](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ClearSelection2 Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ClearSelection2 Method (IModelDoc2)

Clears the selection list.

Visual Basic Syntax

Visual Basic (Declaration)

```
Sub ClearSelection2( _  
    ByVal All As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim All As System.Boolean  
  
instance.ClearSelection2(All)
```

C#

```
void ClearSelection2(  
    System.bool All  
)
```

C++/CLI

```
void ClearSelection2(  
& System.bool All  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

All

True clears the entire existing selection list, false clears only the items in the active selection list (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ClearSelection2](#).

Example

[Add Component and Mate \(C++\)](#)
[Get Bodies \(C++\)](#)
[Traverse Bodies \(C++\)](#)
[Add Distance Mates \(VBA\)](#)
[Create Plane Thru 3 Points In-context \(VBA\)](#)
[Get Names of Bodies in Multibody Part \(VBA\)](#)
[Insert a Note \(VBA\)](#)
[Create Equation-driven Curve \(C#\)](#)
[Create Equation-driven Curve \(VB.NET\)](#)
[Create Equation-driven Curve \(VBA\)](#)

Remarks

False only works if the current PropertyManager page contains a selection list; otherwise, this method clears all selections.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[ISelectionMgr::SuspendSelectionList Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ClearUndoList Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ClearUndoList Method (IModelDoc2)

Clears the undo list for this model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub ClearUndoList()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.ClearUndoList()
```

C#

```
void ClearUndoList()
```

C++/CLI

```
void ClearUndoList();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ClearUndoList](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::EditRedo2 Method](#)
[IModelDocExtension::FinishRecordingUndoObject Method](#)
[IModelDocExtension::StartRecordingUndoObject Method](#)
[IModelDoc2::SketchUndo Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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Close Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : Close Method (IModelDoc2)

Not implemented. Use [ISldWorks::CloseDoc](#).

.NET Syntax

Visual Basic (Declaration)

```
Sub Close()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
  
instance.Close()
```

C#

```
void Close()
```

C++/CLI

```
void Close();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::Close](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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CloseFamilyTable Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : CloseFamilyTable Method (IModelDoc2)

Closes the design table currently being edited.

.NET Syntax

Visual Basic (Declaration)

```
Sub CloseFamilyTable()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.CloseFamilyTable()
```

C#

```
void CloseFamilyTable()
```

C++/CLI

```
void CloseFamilyTable();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::CloseFamilyTable](#).

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IModelDoc2::GetDesignTable Method](#)
[IModelDoc2::DeleteDesignTable Method](#)
[IModelDoc2::IGetDesignTable Method](#)
[IModelDoc2::InsertFamilyTableEdit Method](#)
[IModelDoc2::InsertFamilyTableNew Method](#)
[IModelDoc2::InsertFamilyTableOpen Method](#)
[IModelDocExtension::HasDesignTable Method](#)
[IDesignTable Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

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ClosePrintPreview Method (IModelDoc2)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ClosePrintPreview Method (IModelDoc2)

Closes the currently displayed Print Preview page for this document.

.NET Syntax

Visual Basic (Declaration)

```
Sub ClosePrintPreview()
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
instance.ClosePrintPreview()
```

C#

```
void ClosePrintPreview()
```

C++/CLI

```
void ClosePrintPreview();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ClosePrintPreview](#).

Remarks

Use this method after using [IModelDoc2::PrintPreview](#) to close a Print Preview page.

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ClosestDistance Method (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ClosestDistance Method (IModelDoc2)

Calculates the minimum distance between the specified geometric objects.

.NET Syntax

Visual Basic (Declaration)

```
Function ClosestDistance( _  
    ByVal Object1 As System.Object, _  
    ByVal Object2 As System.Object, _  
    ByRef Point1 As System.Object, _  
    ByRef Point2 As System.Object _  
) As System.Double
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2  
Dim Object1 As System.Object  
Dim Object2 As System.Object  
Dim Point1 As System.Object  
Dim Point2 As System.Object
```

```
Dim value As System.Double
```

```
value = instance.ClosestDistance(Object1, Object2, Point1, Point2)
```

C#

```
System.double ClosestDistance(  
    System.object Object1,  
    System.object Object2,  
    out System.object Point1,  
    out System.object Point2  
)
```

C++/CLI

```
System.double ClosestDistance(  
& System.Object^ Object1,  
& System.Object^ Object2,  
& [Out] System.Object^ Point1,  
& [Out] System.Object^ Point2  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Object1

Pointer to first object (see **Remarks**)

Object2

Pointer to second object (see **Remarks**)

Point1

Array of x, y, z coordinates for the point on Object1 that is nearest to Point2

Point2

Array of x, y, z coordinates for the point on Object2 that is nearest to Point1

Return Value

Distance in meters between Point1 and Point2; -1.0 if no solution

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ClosestDistance](#).

Example

[Calculate Closest Distance Between Faces \(VBA\)](#)

[Calculate Closest Distance Between Model Components \(VBA\)](#)

Remarks

Supported types of Object1 and Object2 include [swSelectType_e](#):

- swSelFACES (face)
- swSelEDGES (edge)
- swSelVERTICES (vertex)
- swSelSKETCHSEGS (sketch segment)
- swSelDATUMPLANES (reference plane)
- swSelEXTSKETCHPOINTS (point on origin)
- swSelDATUMAXES (reference axis)
- swSelCOMPONENTS (component; multi-body part supported, but not multi-part sub-assembly)
- swSelREFCURVES (reference curve)
- swSelSOLIDBODIES (solid bodies only; surface bodies not supported)

This method has these restrictions for drawings:

- Cannot measure between a sketch entity and a model entity
- Measured sketch entities have to belong to the same sheet
- Model entity measurements are based on the model origin
- Measured object cannot be a temporary geometric entity

See Also

[IModelDoc2 Interface](#)

[IModelDoc2 Members](#)

Availability

ConfigurationManager Property (IModelDoc2)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDoc2 Interface](#) : ConfigurationManager Property (IModelDoc2)

Gets the [IConfigurationManager](#) object, which allows access to a configuration in a model.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ConfigurationManager As ConfigurationManager
```

Visual Basic (Usage)



```
Dim instance As IModelDoc2
Dim value As ConfigurationManager

value = instance.ConfigurationManager
```

C#

```
ConfigurationManager ConfigurationManager {get;}
```

C++/CLI

```
property ConfigurationManager^ ConfigurationManager {
    ConfigurationManager^ get();
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[IConfigurationManager](#) object

Visual Basic for Applications (VBA) Syntax

See [ModelDoc2::ConfigurationManager](#).

Example

[Get Configuration Parameters \(C++\)](#)
[Add Derived Configurations \(VBA\)](#)
[Extract Configuration-specific Parameters \(VBA\)](#)
[Get Dimension Values in All Configurations \(VBA\)](#)
[Get Display Dimensions, GTols, and Surface-finish Symbols \(VBA\)](#)
[Get Names of Bodies in Multibody Part \(VBA\)](#)
[Select Bodies \(VBA\)](#)
[Set All Assembly Components Lightweight or Resolved \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VBA\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(VB.NET\)](#)
[Traverse Assembly at Component and Feature Levels Using Recursion \(C#\)](#)

See Also

[IModelDoc2 Interface](#)
[IModelDoc2 Members](#)
[IConfigurationInterface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0